

1 Configurations to Identify the Actual Cycle

Purpose

In a production using clocks, the display of the current actual cycle of a machine is an important information. The system provides different options to calculate the actual cycle:

- Calculating the actual cycle via CT-MSS
- Calculating the actual cycle via CT-UMPS
- Calculating the actual cycle via direct connection to control systems (provision of variables via control system)
- Calculating the actual cycle on the terminal AIP 8.1/AIP 8.2, if you do not use CT-MSS or CT-UMPS
- Calculating the actual cycle on the server

This document describes the different data collection configurations and calculations options of the actual cycle. This document also describes the configuration options for the display on the shop floor terminal AIP.

You can display the actual cycle on the MOC in the following MOC applications:

- [Workplaces/machines](#)
- [Cycle progression](#).



In addition to the current actual cycle, the shop floor data collection (BDE) provides an actual cycle for the respective operation. The calculation of the operation-related actual cycle is described in the document [OBJECT_MES-Operation_ActualData.pdf](#).

Calculating the actual cycle via CT-MSS

The CT-MSS measures the actual cycle with each clock. To calculate the actual cycle, the system uses the first counter configured with the option "cycle monitoring" (counter with the smallest counter number). The sampling rate (precision) is 100 milliseconds by default. The maximum actual cycle that you can identify is 50 minutes.

For machines with very rapid clocks, the sampling rate can be reduced to 20 milliseconds via the following configuration in the ctaip.ini. The maximum actual cycle that you can identify in this case is 10 minutes.

[MSS-INIT]

MSSZyklusReferenz=2

; sampling rate: 0 = 100ms, 2 = 20 ms, default is 0

The actual cycle is passed from the CT-MSS to the terminal where you can visualize it. The terminal sends the actual cycle (acronym IZY) with each status update (dialog M_AST, usually every 120 seconds) to the server. The server then writes the actual cycle to the database (data field maschinen_status.mstck).

Deactivating the calculation of the actual cycle

You can deactivate the calculation of the actual cycle via MSS. Enter the following configuration in the ctaip.ini in section [MSS-INIT]:

[MSS-INIT]

MSSZyklusBerechnung=OFF

(Note: if the entry is not available, the calculation of the actual cycle is performed)

In this case, no actual cycle is passed to the PCC; the PCC calculates the actual cycle (see section [Calculating the actual cycle via PCC](#)).

Calculating the actual cycle via CT-UMPS

The CT-UMPS measures the actual cycle with each clock. To calculate the actual cycle, the system uses the first counter configured with the option "cycle monitoring" (counter with the smallest counter number). The sampling rate (precision) is 10 milliseconds by default.

To this end, you must define an OPC variable Z:Z00X in the configuration file opcmpdv.ini:

Z:Z00X=CT-UMPS/Cycle_Time_Y

The actual cycle is passed from the CT-UMPS to the terminal where you can visualize it. The terminal sends the actual cycle (acronym IZY) with each status update (dialog M_AST, usually every 120 seconds) to the server. The server then writes the actual cycle to the database (data field maschinen_status.mstck).

Deactivating the calculation of the actual cycle

You can deactivate the calculation of the actual cycle via CT-UMPS. Enter the following configuration in the ctaip.ini in section [MSS-INIT]:

[MSS-INIT]

MSSZyklusBerechnung=OFF

(Note: if the entry is not available, the calculation of the actual cycle is performed)

Other option – for example, if the PCC is run stand-alone, the configuration definition Z:Z00X is not performed.

In this case, no actual cycle is passed to the PCC; the PCC calculates the actual cycle (see section [Calculating the actual cycle via PCC](#)).

Calculating the actual cycle via direct connection of control system

The actual cycle can be passed directly from the machine control system as separate variable (e.g. OPC).

To this end, you must define a variable Z:Z00X in the respective configuration file. Examples:

- opcmpdv.ini:
Z:Z001=S7/Cycle_Time
Note: The mapping to the S7 address is made in the OPC server configuration, e.g.
S7/Cycle_Time=DB100.DBW2
- e63.ini:
Z:Z001=ActTimCyc

The actual cycle must be passed in [msec]. The data format of the variable defines the maximum actual cycle.

Deactivating the calculation of the actual cycle

If the machine control cannot calculate an actual cycle, the configuration definition is not performed. In this case, the PCC can calculate the actual cycle (see section [Calculating the actual cycle via PCC](#)).

Calculating the actual cycle on the terminal AIP 8.1

If you do not use CT-MSS or CT-UMPS, the terminal can identify the actual cycle.

Only with terminals in operation mode "MDE", the actual cycle is calculated. The actual cycle is only updated if the machine is in the "production" status and if a valid shift is currently available for the machine according to the defined shift calendar. In addition, the terminal must actually record cycles and data must be transferred from the terminal to the server (online). Otherwise, the previous value is still displayed, even if the machine is no longer in the "production" status.

The system does not calculate the actual cycle, if the quantities are only recorded manually.

The terminal can only identify an actual cycle if at least two clocks are recorded. The first clock after a malfunction is skipped, as otherwise a wrong actual cycle would be identified.

To calculate the actual cycle, the system uses the first counter configured with the option "cycle monitoring" (counter with the smallest counter number). This counter must also be set if operating signals are monitored and the terminal is supposed to identify actual cycles in addition to operating signals.

To smooth cycle time fluctuations in production, the system calculates mean values using the last 20 measured values.

Make the following entries in the configuration file ctaip.ini on the Windows terminal:

[MSS-INIT]

CalculateCycle=ON

MSSZyklusberechnung=OFF

If CalculateCycle=ON, then MSSZyklusBerechnung=ON is implicitly disabled, if available.

[Layout]

ActualCycleView=on

→The terminal sends the actual cycle (acronym IZY) with each status update (dialog M_AST, usually every 120 seconds) to the HYDRA server. The HYDRA server then writes the actual cycle to the database (data field maschinen_status.mstck).

Calculating the actual cycle via PCC / on the terminal AIP 8.2

The PCC calculates the actual cycle, if the following conditions are fulfilled:

- The PCC is active, i.e. the PCC runs either stand-alone or in an AIP 8.2 in so-called combined operation.
- The file mdeb2.dll has at least version 8.1.0.16.
You can find the version number in file #dirinfo.lst in the spool directory. The file is generated with a terminal upload. To start the terminal upload, use the function *Terminal administration > Request diagnosis files* in the [Terminal configuration](#).
- Via CT-MSS, CT-UMPS or via OPC, no actual cycle is passed, i.e.
 - no Z channel is configured or
 - in the ctaip.ini, MSSZyklusberechnung=OFF is configured
- In the [Counter configuration](#), at least one counter is configured where the option *Cycle monitoring* is enabled.

To calculate the actual cycle, the system uses the first counter configured with the option "cycle monitoring" (counter with the smallest counter number). This counter must also be set if operating signals and actual cycles are identified. The setting *Monitoring type* in the [Workplace and resource configuration](#) has no effect on the calculation of the actual cycle.

If the option *Cycle monitoring* is not enabled with any counter, the actual cycle is not calculated. No acronym IZY is passed to the server.

The actual cycle is only identified, if the machine is in *production*. And the system must actually record cycles (via C channels). Otherwise, the value 0 is passed to the terminal or server as actual cycle. If quantities are only recorded manually, the PCC does not calculate an actual cycle.

To identify an actual cycle, at least two clocks must have been recorded. The first clock after a malfunction is skipped, as otherwise a wrong actual cycle would be identified.

Calculation of the mean value

To smooth cycle time fluctuations in production, the system calculates mean values using the last 20 values of actual cycles. See the following examples:

For a better overview, the actual cycle is presented in seconds in the examples that follow (e.g. 2.1 seconds). Internally, the system identifies and saves the actual cycle in milliseconds (e.g. 2119 [msec]).

The system uses the last 20 actual cycles remembered and the current actual cycle (curr.) to calculate the mean value for the actual cycle that is sent to the terminal or server. At the same time, the current actual cycle is remembered (not the mean value); the oldest actual cycle that has been remembered is removed.

Time t0:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
2.0	2.1	2.0	2.0	2.1	2.1	2.1	1.9	2.0	2.1	2.0	2.0	2.0	2.1	2.0	2.1	2.1	2.0	2.1	2.1	2.0

20...1 = the last 20 actual cycle values identified | curr. = currently identified actual cycle

Time t1:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
2.1	2.0	2.0	2.1	2.1	2.1	1.9	2.0	2.1	2.0	2.0	2.0	2.1	2.0	2.1	2.1	2.0	2.1	2.1	2.0	1.9

If an actual cycle is identified that deviates significantly from the actual cycle values previously recorded, this actual cycle is "channelled" through the 20 values. This deviating actual cycle affects the mean value displayed for the actual cycle during this period of time.

Time t0:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
2.0	2.1	2.0	2.0	2.1	2.1	2.1	1.9	2.0	2.1	2.0	2.0	2.0	2.1	2.0	2.1	2.1	2.0	2.1	2.1	9.0

Time t7:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
2.0	2.1	2.0	2.0	2.1	2.1	1.9	2.1	2.0	2.1	2.0	2.0	2.0	9.0	2.0	2.1	2.1	2.0	2.1	2.1	2.0

Time t14:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
2.1	2.0	2.1	2.0	2.0	2.0	9.0	2.0	2.1	2.1	2.0	2.1	2.1	2.0	1.9	2.0	2.1	2.0	2.0	2.1	1.9

In case of a malfunction, all previously recorded values for the actual cycle are deleted:

Last calculation of mean value before changing from production → malfunction:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
2.0	2.1	2.0	2.0	2.1	2.1	2.1	1.9	2.0	2.1	2.0	2.0	2.0	2.1	2.0	2.1	2.1	2.0	2.1	2.1	9.0

Change production → malfunction:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.

After a restart, i.e. after a change from malfunction → production, no values for the actual cycle of the past are available. The calculation of the mean value therefore uses only the values for the actual cycle remembered up to the respective point in time, until 20 values are again available for the actual cycle:

Time *t0* after change from malfunction → production:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
																				1.9

Time *t6*:

20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	curr.
														1.9	2.0	2.0	1.9	2.1	2.1	2.0

Note: You can change the number of measured values used to calculate the mean value. To change the number, you must explicitly request the acronym MWANZ for the [machine list] in the configuration file ctaipplay.ini. Also specify the *Number of cycles to be evaluated* (1 to a maximum of 9) in the [Workplace configuration](#) on the MOC. If you enter the value 0 in the input field, the default number of 20 measured values is used.

Communication with terminal AIP 8.2 or server

The mean value of the actual cycle (acronym IZY) is cyclically sent – every 120 seconds – to the server as part of the so-called status update (dialog M_AST). And also with each status change (dialog M_MST), the actual cycle is sent to the server. The HYDRA server then writes the actual cycle to the database (data field maschinen_status.mstck).

If the status changes from production to → malfunction, the status update (M_MST) includes the last actual cycle of the production.

Display of the actual cycle on the terminal if the MDE data collection runs in combined operation

The actual cycle on the terminal AIP 8.2 is immediately updated.

Display of the actual cycle on the terminal if the MDE data collection is performed with a stand-alone PCC

The actual cycle is updated on the terminal AIP 8.2 with the cyclic reload of the machine list (default: every 10 minutes).

Display of the actual cycle on the MOC

If communication to the server is not possible, the MOC continues to show the last value. This value is also shown if the machine is not in production any more.



The actual cycle displayed on the MOC can be different to the actual cycle displayed on the AIP 8.2 because the actual cycle is only sent to the server in case of specific events (dialogs M_AST, M_MST).

Identification of actual cycle on the server

If the actual cycle cannot be calculated on the terminal or via PCC and if the following requirements are fulfilled, then the server calculates the actual cycle:

- The dialog string that the terminal/PCC sends to the server does not include the acronym IZY.
- In the [Counter configuration](#), the option "Posting as cycles" is enabled for at least one counter.

The server uses the following formula to calculate the actual cycle:

$$\text{Actual cycle} = \frac{\text{Duration}}{\text{Cycles recorded during status "Production (AGR:HUB)"}}$$

The time results from the difference between current time and the time when the actual cycle was last calculated, i.e. since the last status update (dialog M_AST, acronym "IZY", usually every 120 seconds).



The server might calculate an actual cycle, which does not necessarily match the actual cycle time of the machine!

Display of cycles on the terminal AIP 8.1

Via customization, you can display the actual cycle/target cycle in the dialogs on the AIP that show machine information (MMINFO and MINFO). The following formats are supported:

- Seconds/1000 cycles (equals milliseconds per 1 cycle)
- Seconds/cycle
- Cycles/minutes

You must store the acronym MNR.IZY in the respective dynamic dialogs to this end.

In AIP standard processing, the field is automatically downloaded when data is reloaded.



If you use a customized version of the ctaipplay.ini file, this file could overwrite the downloaded field.

To display the respective format, you must store the required acronym in the dialogs:

- Display "actual cycle [sec/cycle] / [sec/stroke]" using "MNR.IZY/HUB"
 - Acronym: **MNR.IZY/HUB** referred to as **Actual cycle [sec/cycle]** or **Actual cycle [sec/stroke]**
- Display "actual cycle [cycles/min.] / [strokes/min.]" using "MNR.IZY/MIN"
 - Acronym: MNR.IZY/MIN referred to as **Actual cycle [cycles/min.]** or **Actual cycle [strokes/min.]**

Display of cycles on the terminal AIP 8.2

The document [AIP2 Configuration GUI.pdf](#) describes in chapter *Constants (Defines)*, section *FORMAT_CYCLE*, and in chapter *CalculatedFields* the configuration options that are available to display the actual cycle or target cycle in the tile view on the AIP 8.2.

To display the actual cycle or the target cycle in the dynamic dialogs, e.g. in the machine information, the configuration options described in the previous section apply (*Display of cycles on the terminal AIP 8.1*).